How-To Library

Informatica

Configuring Data Domains for Unstructured Sources in Data Privacy Management

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Abstract

You can use predefined data domains that are preconfigured with domain modules or configure custom data domains to perform data discovery on unstructured sources using the Informatica Discovery Agent. This article describes about domain modules, module file configuration, list of predefined data domains, and custom data domains.

Supported Versions

- Data Privacy Management 10.4.1

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Overview

The Data Privacy Management installation includes a set of commonly used, predefined data domains that you can use to identify and protect sensitive data in a data store. You can also create custom data domains. Data Privacy Management supports 71 predefined data domains that are preconfigured with domain modules to use with the Informatica Discovery Agent to discover data for unstructured sources.

During the data discovery process, Data Privacy Management provides the details of the unstructured sources and data domain configuration to the agent. The configuration parameters differ for data domains when the agent is used for data discovery. For data domains, you can provide parameters such as keywords, multiple regular expressions, and confidence levels, to perform data discovery using the agent.

Domain Modules

Each supported data domain has a preconfigured domain module. You can also configure custom data domain modules. When you configure domain modules, the name of the domain module must be same as the custom data domain name.

Each domain module contains the following files:

RuleConfig.json file

The file that contains the definition of the data domain.

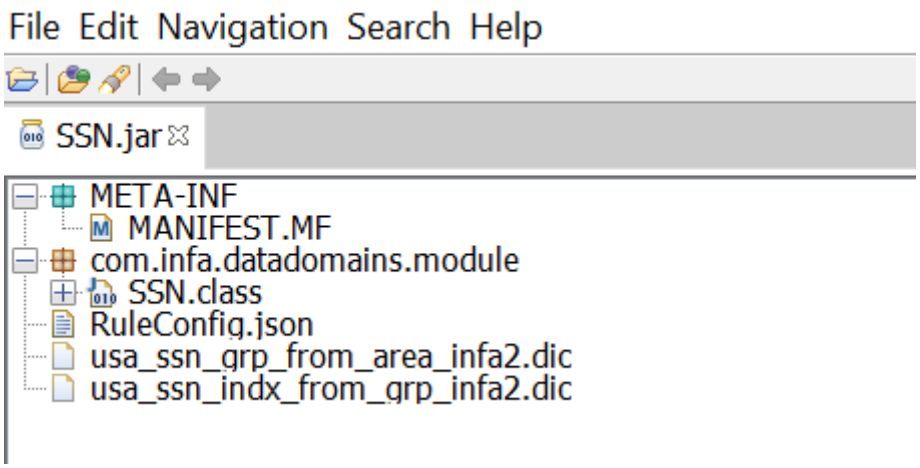
Java class file

The file that contains the verification logic.

Dictionary files

The file for reference table lookup.

The following image shows a sample domain module content:



In the example, you can view the following content in the SSN.jar module:

- The Java class file: `SSN.class`
- The data domain configuration file: `RuleConfig.jscon`
- The dictionary files for reference table lookup: `usa_ssn_grp_from_area_infa2.dic` and `usa_ssn_indx_from_grp_infa2.dic`

Domain Module Location

The `RuleConfig.json` file is bundled in the `.jar` module file in the following location:

```
<DPM_10.4.1_INSTALL>/SecureAtSourceService/accelerators
```

Rule Configuration File

The `RuleConfig.json` file is the configuration file that contains the definition of a data domain.

Use the following syntax to configure the `RuleConfig.json` file:

```
RuleConfig: {
"Name": <String>, //Mandatory
"ReferenceTables": <List<String>>, //Optional
"Regexes": [{                    //Optional
          "ConfidenceLevel": <String>,
          "Pattern": <String>,
          "Description": <String>
        }],
 "FullyQualifiedClassName" : <String>      // Optional
"KeyWords": <List<String>>          //Optional
 }
```

The following table describes the properties of the `RuleConfig.json` file.

Property	Description
Name	Required. Name of the data domain.
ReferenceTables	Optional. List of the reference tables and dictionaries for lookup. Informatica recommends not to modify the settings. **Note:** You need to provide values for in this field if you have not provided the regular expressions.

Property	Description
Regex	Optional. Multiple regular expressions can be defined with the corresponding confidence level. You can use more than one regular expression to match against the data. You can enter the following confidence level values: High, Medium, or Low. **Note:** You need to provide values for in this field if you have not provided the reference table values.
FullyQualifiedClassName	Optional. Fully qualified class name for the main class of the verification module. This class is used to verify the data domain. **Note:** You cannot add additional verification classes other than the predefined verification classes.
KeyWords	Optional. List of keywords. The keywords are used to boost the confidence level for a particular token.

Sample Rule Configuration for SSN

The following image shows a sample configuration with regular expressions for the SSN data domain in the `RuleConfig.json` file:

RuleConfig.json

```
RuleConfig: {
     "Name": "SSN",
     "ReferenceTables": [],
     "Regexes": [
          {
               "Pattern": "\\b(([0-9]{3})[ -]?([0-9]{2})[ -]?([0-9]{4}))\\b",
               "ConfidenceLevel": "High"
          }
     ],
     "FullyQualifiedClassName":"com.infa.datadomains.module.SSN"
}
```

For the SSN data domain, the Regex section shows a regular expression pattern and confidence level.

Sample Rule Configuration for City

The following image shows a sample configuration with reference tables for the City data domain in the `RuleConfig.json` file:

RuleConfig.json

```
RuleConfig: {
      "Name": "City",
      "KeyWords": [],
      "ReferenceTables": [{"Name": "cities_of_world_infa",
                           "Value" : "cities_of_world_infa.dic"}],
      "Regexes": [
            {
                  "Pattern": "",
                  "ConfidenceLevel": ""
            }
      ],
      "FullyQualifiedClassName":"com.infa.datadomains.module.City"
}
```

For the City data domain, the Reference Tables section shows a name and value.

Modify Predefined Data Domains

You can use predefined data domains that are supported to use with the Informatica Discovery Agent for data discovery on unstructured sources. For the list of supported data domains, refer to "Supported Predefined Data Domains" on page 8. For more information on using the agent with unstructured resources, see the *Informatica Data Privacy Management 10.4.1 User Guide*.

If you want to modify any predefined data domain configuration, such as keywords, you can extract the respective module jar and make changes to the configuration in the `RuleConfig.json` file.

Informatica recommends that you do not change any predefined data domain configuration. Instead, you can clone the data domain that you want to modify. Create a copy of the corresponding module jar file and rename the copied module jar to the cloned data domain name before you update the configuration in the `RuleConfig.json` file.

Note: Any changes that you make to the predefined data domain modules will be lost after you upgrade Data Privacy Management. It is recommended not to make changes to the predefined domain modules.

Custom Data Domains

You can configure and use a custom data domain with the Informatica Discovery Agent to discover data for unstructured sources.

You can use the regular expression pattern-based matches for custom data domains to perform data discovery on unstructured sources. You cannot use the Reference Table and Rule methods for custom data domains.

To perform data discovery on unstructured sources using the agent, you can use one of the following methods to configure custom data domains based on your requirement:

- Configure custom data domains with only regular expression based matches. You can use a custom domain that is created through the Data Privacy Management user interface.

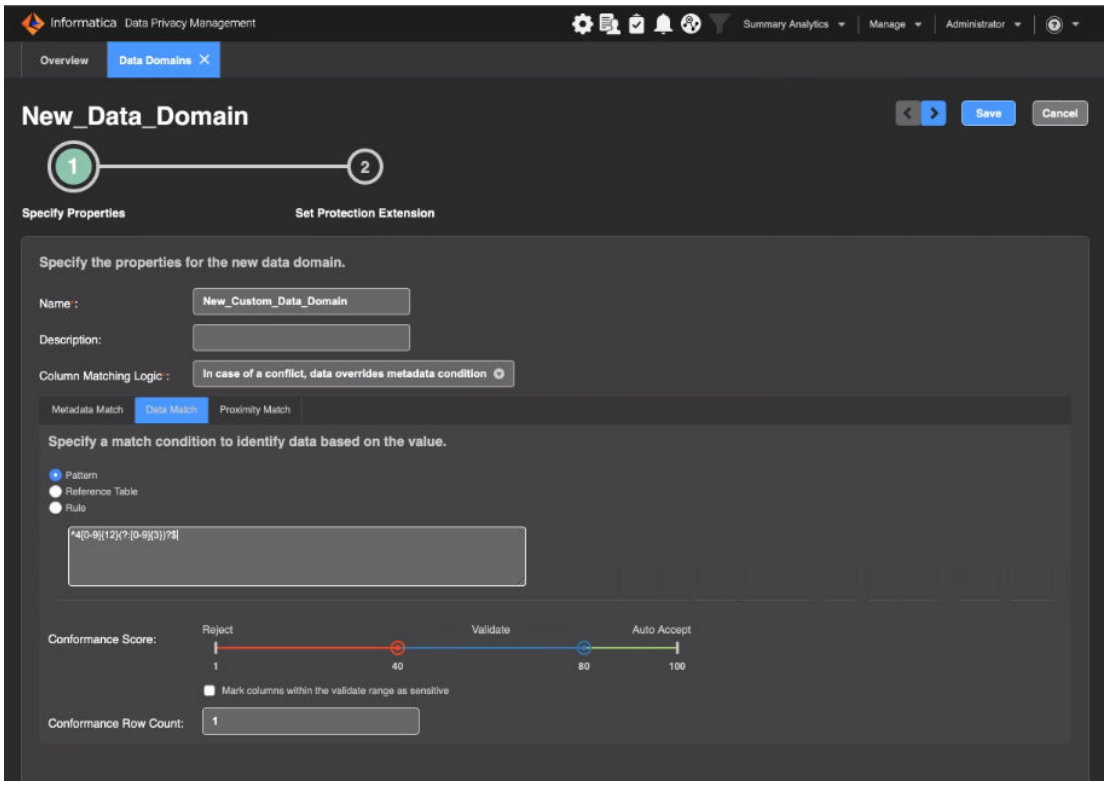
- Configure custom data domains with regular expressions and advanced match configuration, such as keywords. You can create a custom data domain from the Data Privacy Management user interface and create the corresponding domain module in the domain module location.

Note: You cannot use the Metadata Match conditions.

Regular Expression Pattern

If data discovery on unstructured sources is based only on a regular expression pattern, you can configure the custom data domain modules through the Data Privacy Management user interface. You can choose the Pattern method in the Data Match condition and enter the regular expression. The configuration details are passed to the Informatica Discovery Agent when the data domain is used in the policy while scanning the unstructured sources.

The following image shows a sample pattern-based data domain:



Regular Expression Pattern with Advanced Configuration

If data discovery on unstructured sources is based on regular expressions and advanced configuration, such as keywords, you have to configure rule-based custom data domains from the Data Privacy Management user interface.

Create a custom data domain of Rule type in the Data Match condition. Create a corresponding domain module for the custom data domain. The configuration of the custom data domain is used from the domain module and not from the rule-based custom data domain on the user interface. The configuration details are passed to the Informatica Discovery Agent when the data domain is used in a policy while scanning the unstructured sources.

Creating a Custom Data Domain with Advanced Configuration

Create a module for custom data domains to use with the agent for the discovery process. You can provide a regular expression with additional details, such as keywords, for a rule-based custom data domain.

1. Create a custom module that conforms to the domain modules conditions.

   For more information, refer to "Domain Modules" on page 2.

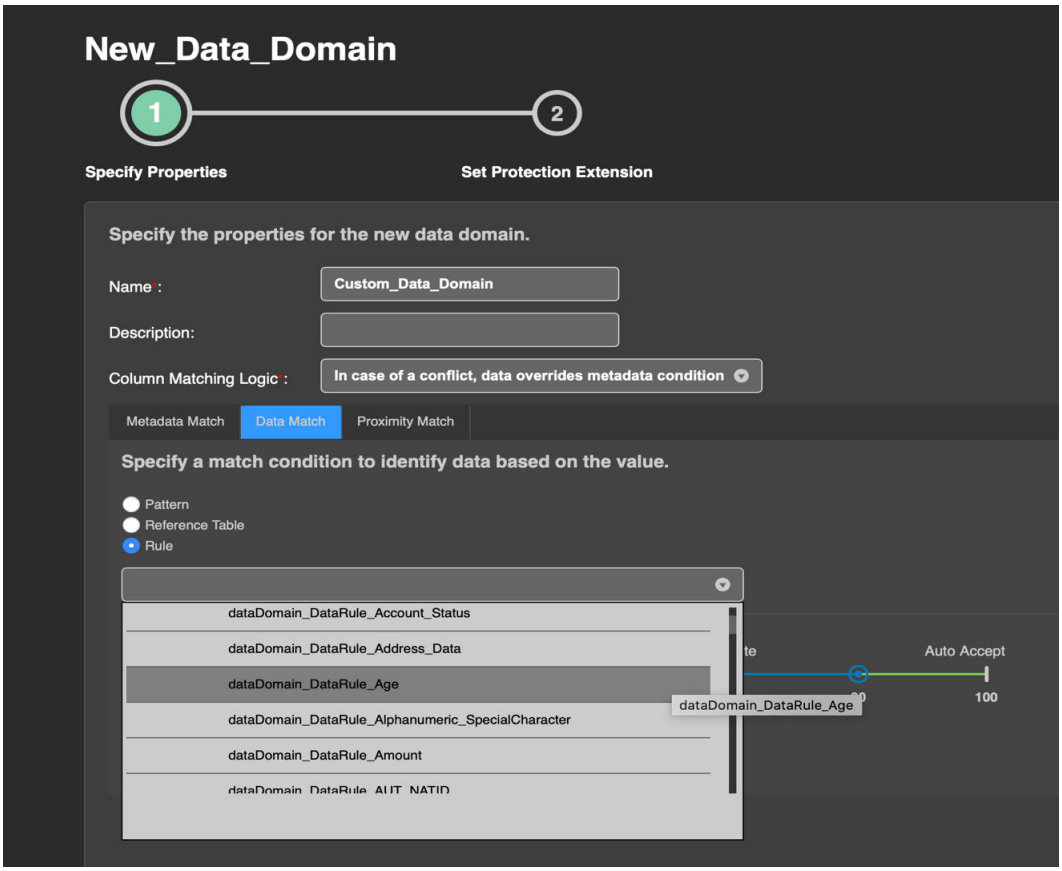
2. Place the domain module in the following location:

   ```
   <DPM_10.4.1_INSTALL>/SecureAtSourceService/accelerators
   ```

   You can also create a copy of an existing module and update the configuration. Refer to "Modify Predefined Data Domains" on page 5

3. Create any rule-based custom data domain from the Data Privacy Management user interface. Or clone an existing predefined data domain that is rule based.

   Ensure that the name of the custom data domain is same as the domain module name.
   The following image shows a sample rule-based custom data domain:



You can use the configured custom data domain with the Informatica Discovery Agent when you perform data discovery on unstructured sources. For more information on using the agent with unstructured resources, see the *Informatica Data Privacy Management 10.4.1 User Guide*.

Supported Predefined Data Domains

The following table describes the predefined data domains that Data Privacy Management supports for data discovery on unstructured sources:

Domains	Policy	Lookup Based
AccountNumber	PCI, PHI	No
Address	PHI	No
Admission_date	PHI	No
Age	PHI, PII	No
Austria_NationalID	--	No
Bank_Routing_Number_ABA_Number	--	No
BIC_SWIFTCode	--	Yes
BirthDay	PII, PHI	No
BirthPlace	PII, PHI	No
Canada_Driver_License_Narrow	--	No
Canada_SIN	--	No
CertificateLicenseNumber	GDPR, PHI	No
City	GDPR, PHI	Yes
CompanyName	GDPR, PII	Yes
Computer_Address	GDPR	No
Country	GDPR, PII, PHI	Yes
CountryCode_Phone	GDPR	Yes
CreditCardNumber	PCI, PII	No
Date_AllFormats	--	No
Denmark_NationalID	--	No
DeviceSerialNumber	PHI	No
Discharge_date	PHI	No
DriverLicense_Canada	--	No
DriverLicense_US_Canada_GBR	--	No
DriversLicense_GBR	--	No

Domains	Policy	Lookup Based
DriversLicense_USA	PHI	No
DriversLicense_USA_Narrow	PHI	No
DrivingLicenseNumber	PII	No
Dutch_BSN_NIK	--	No
Email	PHI, PII	No
ExpirationDate	PCI	No
FirstName	PCI, PHI, PII	Yes
France_INSEE	--	No
FullName	--	Yes
Gender	GDPR, PII, PHI	No
Geocode_Latitude	GDPR	No
Geocode_LatitudeLongitude	GDPR	No
Geocode_Longitude	GDPR	No
Grade	GDPR, PII	No
GreatBritain_NINO	--	No
Health_Plan_Beneficiary_Number	GDPR, PHI	No
Height	GDPR, PHI	No
Hostname	GDPR	No
IBAN	GDPR	No
ICD_10_Codes	GDPR, PHI	Yes
ICD_9_Codes	GDPR, PHI	Yes
IPAddress	GDPR, PII	No
Italy_FiscalCode	--	No
JobPosition	GDPR, PII, PHI	Yes
LastName	GDPR, PCI, PII, PHI	Yes
MaidenName	--	Yes
MiddleName	GDPR, PCI, PII, PHI	Yes

Domains	Policy	Lookup Based
NationalID_EMEA	--	No
Norway_NationalID	--	No
Passport_DEU_MR	GDPR	No
Passport_GBR	GDPR	No
Passport_MachineReadable	GDPR	No
Passport_USA_MR	--	No
PhoneNumber	GDPR, PHI	No
Postcode	GDPR, PCI	No
Race	GDPR, PHI	Yes
Religion	GDPR, PHI	Yes
Salary	GDPR, PII	No
SSN	PHI	Yes
SSN_General	PII	Yes
State	GDPR, PII	Yes
Street	GDPR, PII	No
Track1_Format_B	--	No
UniqueIdentifyingNumber	GDPR, PHI	No
URL	GDPR, PHI	No
USA_Bank_Account	--	No
USA_National_Drug_Code_NDC	--	Yes
USA_National_Provider_Identifier_Standard_NPI	--	No
USA_Taxpayer_Identification_Number	--	No
USZip_5digit	--	Yes
VehicleRegPlateNumber	GDPR, PHI, PII	No
Weight	GDPR, PHI	No

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